Two-Dimensional Plate Heat Conduction

Aim: To compare Jacobi and line-TDMA/SOR solutions of the steady two-dimensional Laplace equation against an analytical solution.

Case definition and parameters: Simulations were analyzed on the following parameters.

Nx=81, 30

Ny= 61,15

Temperature at source= 100

Max Iterations = 10000

Tolerance = $1x10^{-9}$

Governing Equation:

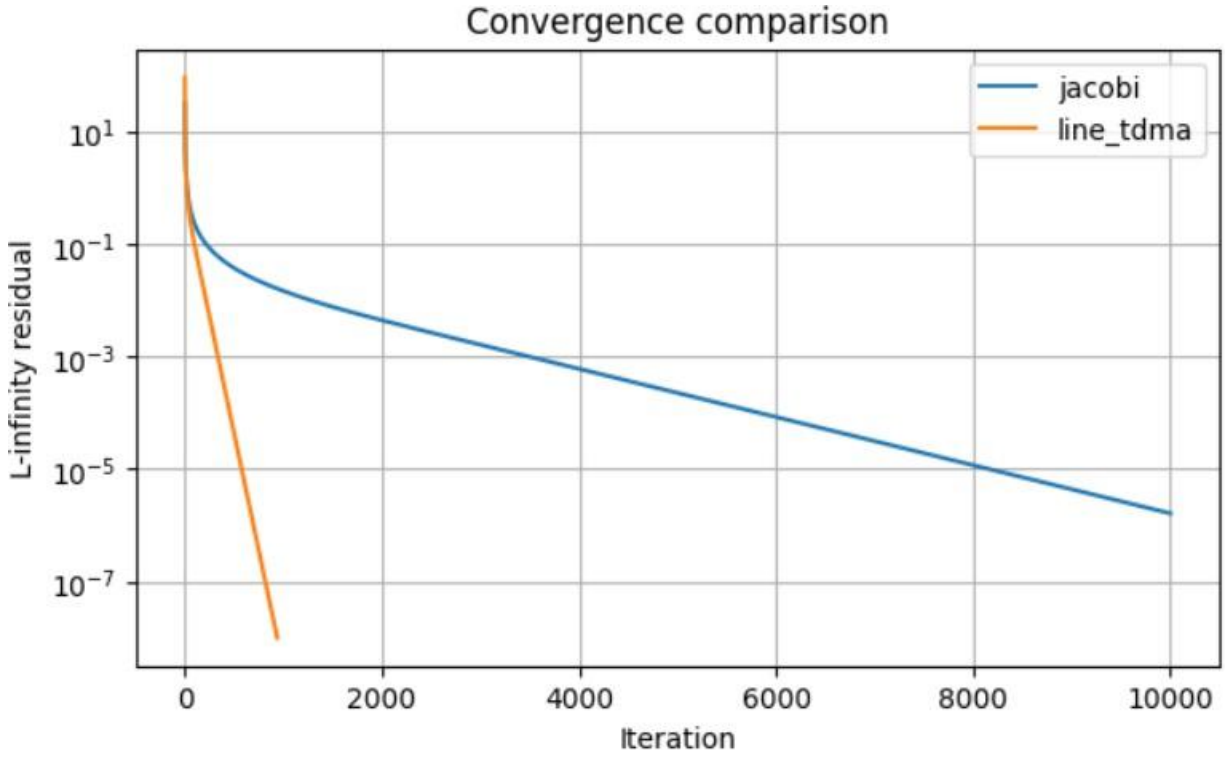
$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0.$$

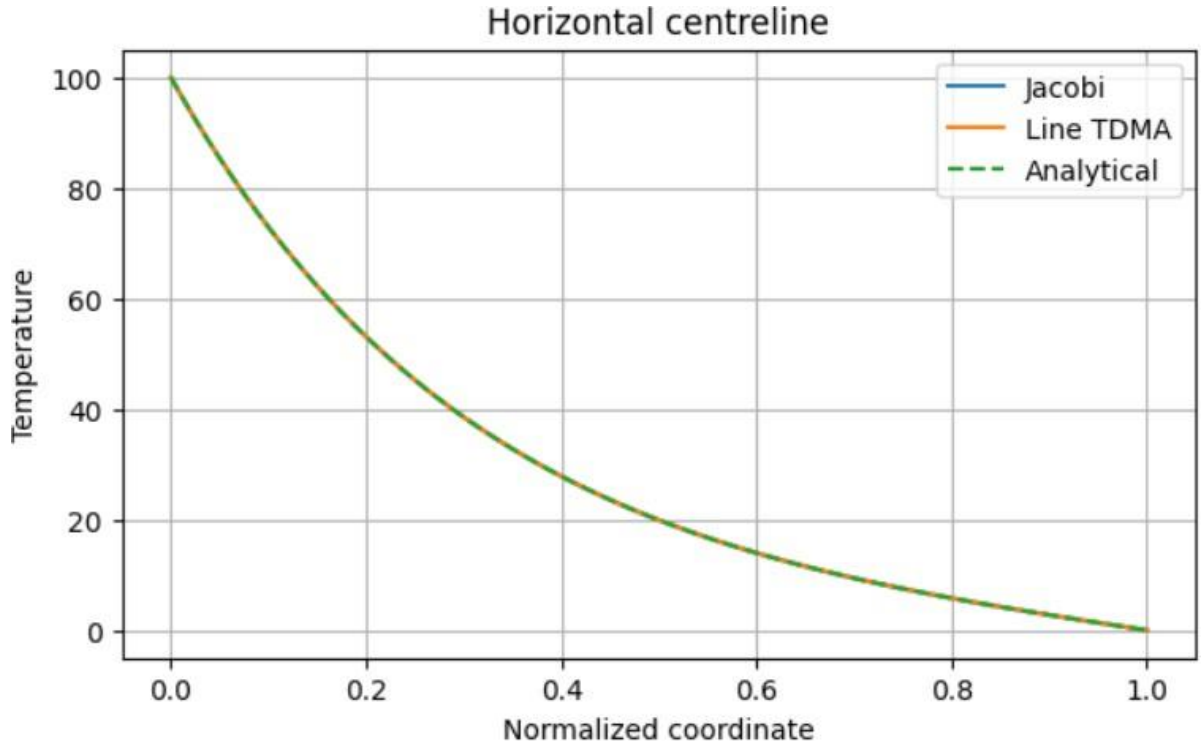
Boundary Conditions: $T(0,y) = T_L \sin(\pi y/H)$; the right, top, and bottom boundaries are held at zero temperature.

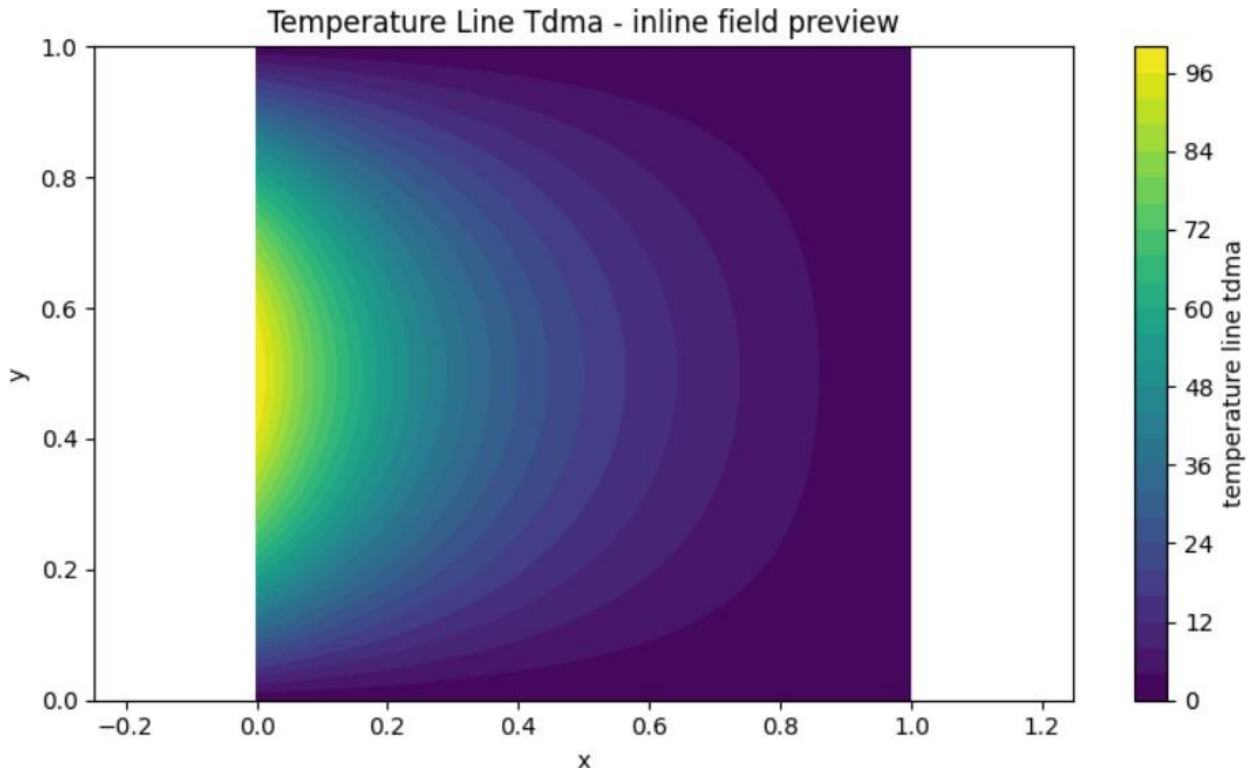
Numerical Method: Second-order finite differences with point Jacobi iteration and line-implicit TDMA/SOR iteration.

Observations: Iteration counts, final residuals, relative L2 errors, center line profiles, and influence of relaxation factor and grid size. Compare both methods on at least two grids. Report convergence rate, analytical error, and the effect of SOR relaxation factor.

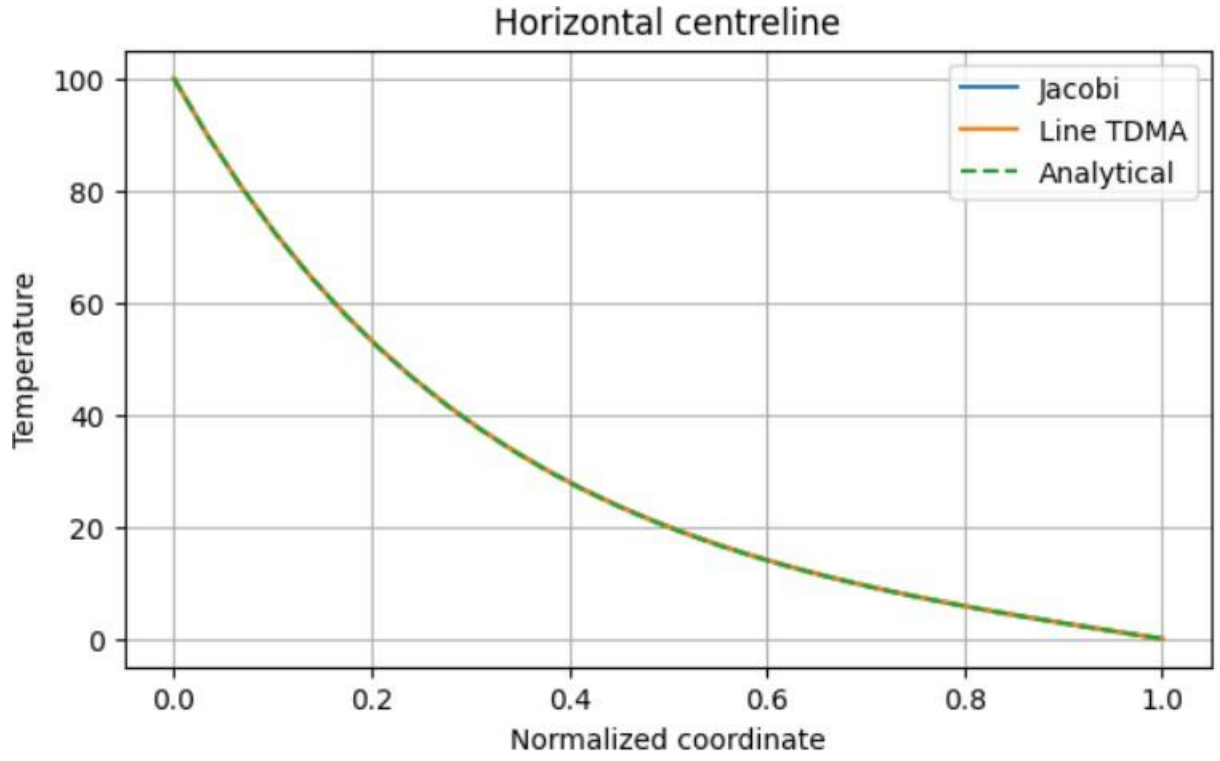
For nx=81 and ny=61

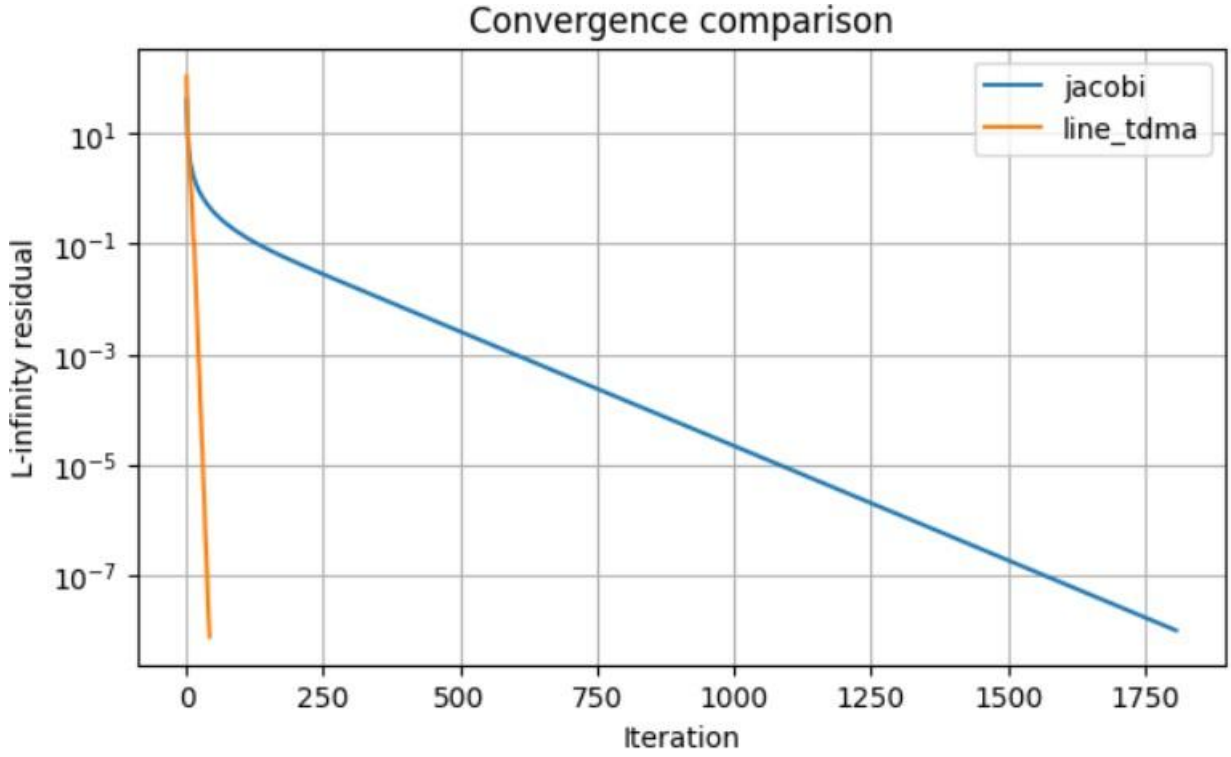


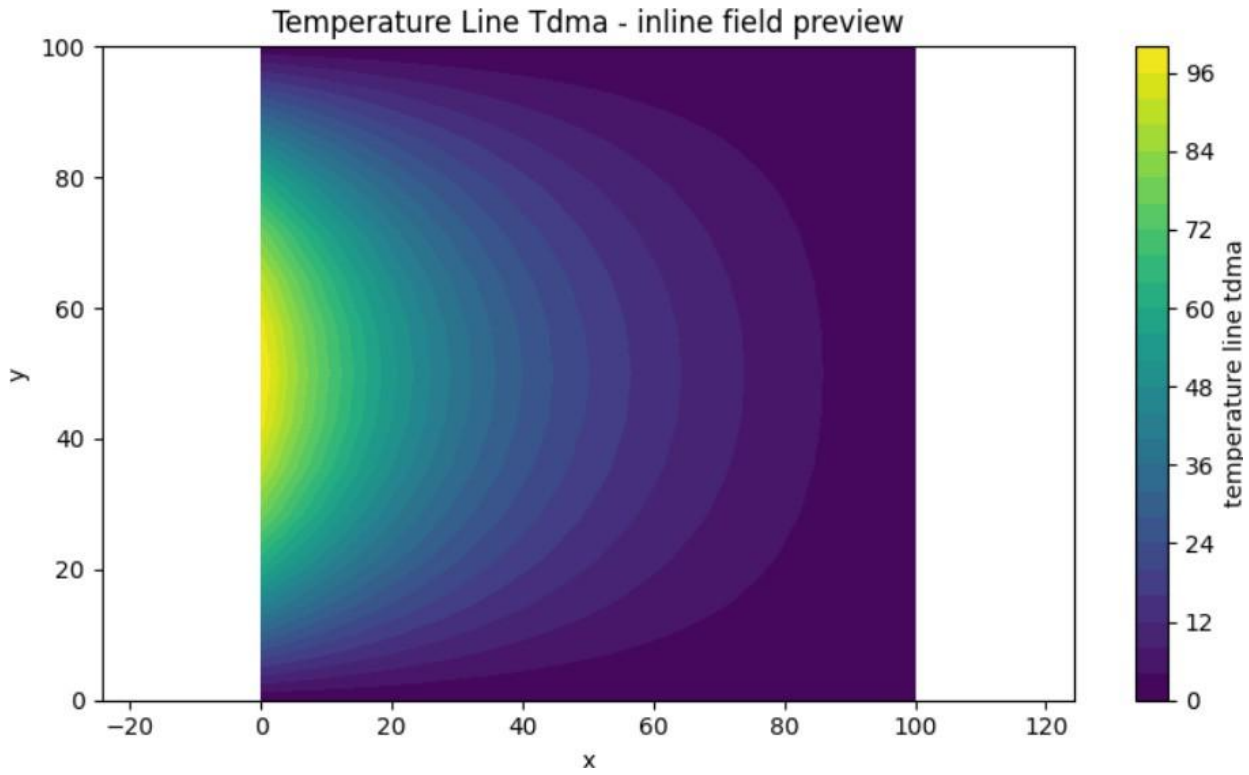




For nx=30 and ny=15


Horizontal centreline
Jacobi
Line TDMA
Analytical
Temperature
Normalized coordinate
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0


Convergence comparison
jacobi
line_tdma
L-infinity residual
Iteration
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
0
250
500
750
1000
1250
1500
1750



Results:

Parameter	**Fine Grid**	**Coarse Grid**
Grid size, N_x x N_y	81x61	30x15
Source temperature	100	100
Maximum iterations allowed	10,000	10,000
Convergence tolerance	10^{-9}	10^{-9}
Jacobi iterations shown	10,000	1,800
Line-TDMA iterations to reach tolerance	800–900	200

Jacobi final L_{infty} residual	$2x10^{-6}$	10^{-9}
Line-TDMA final L_{infty} residual	Below 10^{-9}	Below 10^{-9}
Center line profile agreement	Very good	Very good
Relative convergence speed	Slower	Faster

Conclusion:

The two-dimensional steady heat conduction problem was solved using the Jacobi and line-TDMA iterative methods on grids of 81x61 and 30x15. Both methods produced temperature distributions that closely matched the analytical center line solution, with a maximum temperature of approximately 100 at the midpoint of the heated boundary and a gradual decrease toward the zero-temperature boundaries.

The convergence results showed that line-TDMA was substantially more efficient than Jacobi. On the coarse grid, Jacobi required approximately 1,800 iterations to reach the specified tolerance of 10^{-9}, whereas line-TDMA required approximately 200 iterations. On the fine grid, line-TDMA reached the tolerance within approximately 800–900 iterations, while Jacobi remained at a residual of approximately $2x10^{-6}$ after 10,000 iterations.

The results demonstrate that increasing grid resolution increases the computational effort required for convergence, especially for point-Jacobi iteration. The line-TDMA method handles this increase more efficiently because of its line-implicit solution procedure. Therefore, line-TDMA is the preferred method for this problem when faster convergence and satisfaction of a strict residual tolerance are required.